

VIKAS NAGARAJ

Software and Firmware Developer | Bachelor's Degree in Engineering Sciences (Electronics)

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SUMMARY

A passionate and driven professional with over 5 years of diverse experience across industries. Known for a solution-oriented mindset, adaptability, and a fast-learning ability. Excels in Firmware and Electronic Engineering, transforming passion into a thriving career by delivering innovative solutions and overcoming complex challenges.

CERTIFICATIONS

- Prompt Engineering
- Power BI
- Python Programming and API Testing
- Embedded Firmware Development
- Python
- ROS

PROJECTS

- Successfully managed and coordinated seven EU-funded projects in the energy sector, including Phoenix, Hyperide, Nemo, Enershare, DigiBuild, ensuring timely and effective project execution
- Served as Project Manager at Emotion, Perugia, Italy (<https://emotion-team.com/>), overseeing project planning, stakeholder collaboration, and delivery within a European consortium framework
- Cathodic Protection of Gas Lines : Pietro Fiorentini S.P.A: Developed firmware for a device that monitors pressure and detects damage in gas pipelines, triggering user alerts for enhanced safety. Integrated RS-485, Modbus RTU/TCP protocols, and I2C/SPI interfaces for efficient communication and data logging, ensuring system reliability and compliance with industry standards.

COMPETENCES

- Embedded Systems: RS-485, RS-232, Modbus RTU, Modbus TCP, CANbus, I2C, SPI
- Programming Languages: C, C++, Python, Micropython, Embedded C
- Tools & Platforms: Microsoft Azure DevOps, Visual Studio, Git, Jira, STM Cube, Arduino, IAR, IntelliJ, QT
- Operating Systems: Windows, Linux, Mac OS
- Industrial Protocols: Modbus, TCP/IP, Bluetooth

PROFESSIONAL EXPERIENCE

Firmware Developer | Pietro Fiorentini SPA | La Spezia, Italy

Dec 2023 – Present

- Developed firmware for Cathodic Protection of Gas lines, handling communication via RS-485, Modbus RTU, Modbus TCP, and Bluetooth
- Integrated I2C and SPI for flash modules and performed electrical wiring and maintenance
- Collaborated in debugging, writing C and C++ scripts, and testing the system's API

Firmware Engineer | NTS SRL | Genova, Italy

April 2023 – Oct 2023(Contractual)

- Worked on embedded firmware development for air conditioning systems in trains, utilizing Modbus RTU, RS-485, and CANbus for temperature control and power monitoring
- Developed and tested API using Python, with communication over TCP/IP
- Created JSON, QML, and QT scripts for system operations and maintenance

Electronic Engineer | Emotion SRL | Perugia, Italy

April 2021 – Jan 2023

- Developed software using Python for EV charging towers with integration of Modbus TCP/IP protocols
- Hands-on experience in electrical wiring, hardware development, and API testing for Modbus and CanBus protocols
- Participated in hardware development for DC-DC power systems and collaborated on EU projects involving Nissan Leaf CHAdeMO plug

Technical Support Executive | Convergys India Pvt. Ltd | Technical Support for Microsoft Process

April 2016 - October 2016

- Assisting the customers with their computer office and windows related problems
- Troubleshooting Windows and Office softwares over the telephone by taking remote access.
- Activation and Installation of Softwares to the customers devices as per their needs.
- Creating Logs and making notes on each call taken mentioning the assistance provided.

Customer Support Officer | Mphasis India | Banking Assistance Customer Representative for T D Canada Trust

October. 2016 –July 2017

Role: Assisting the customers with their Banking related issues dealing with creditcards and Debit Cards and daily banking transactions.

- Assisting the customers in creating or changing the debit card PIN and credit card Pin.
- Internet Banking related issues providing assistance for the usage and activation.
- Fraud and Non Fraud claims in case of miscellaneous activities on the customer accounts.

EDUCATION

- Bachelor's Degree in Engineering Sciences (Major in Electronics) | Universita 'Degli Studi di Roma Tor Vergata, Rome, Italy | 2017 – 2022
- Diploma in Electrical and Electronics Engineering | Bapuji Polytechnic, Karnataka, India | 2012 – 2016